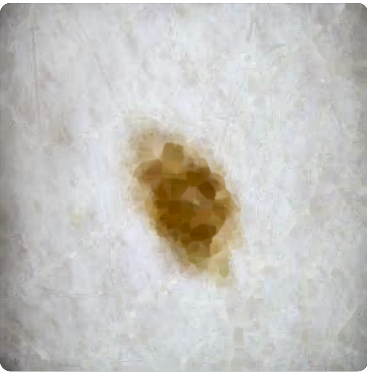


Skin / Melanoma AI Diagnostic Report

Generated: 6/29/2026, 3:02:18 PM · Modality: Dermoscopy · Case #700cc50b · Model: medgemma1.5: RAG-Augmented



OVERALL IMPRESSION

All four images show features highly suspicious for melanoma based on dermoscopic criteria, despite a low malignancy probability from the HAM10000 classifier. Urgent excision is recommended.

LESIONS (4)

ID	SCORE	SIZE	LOCATION
L1	13 7-point dermoscopy	—	Image 1 — <site>
L2	12 7-point dermoscopy	—	Image 2 — <site>
L3	12 7-point dermoscopy	—	Image 3 — <site>
L4	11 7-point dermoscopy	—	Image 4 — <site>

L1

- A

Yes
- B

Yes
- C

Yes
- D

—
- E

—

MAJOR FEATURES

- Atypical pigment network
- Blue-white veil
- Atypical vascular pattern
- Irregular streaks / pseudopods
- Irregular dots / globules
- Irregular blotches
- Regression structures

ANCILLARY FEATURES

- Irregular streaks / pseudopods
- Irregular dots / globules
- Irregular blotches
- Regression structures

L2

A Yes

B Yes

C Yes

D —

E —

MAJOR FEATURES

- Atypical pigment network
- Blue-white veil
- Atypical vascular pattern
- Irregular streaks / pseudopods
- Irregular dots / globules
- Irregular blotches
- Regression structures

ANCILLARY FEATURES

- Irregular streaks / pseudopods
- Irregular dots / globules
- Irregular blotches
- Regression structures

L3

A Yes

B Yes

C Yes

D —

E —

MAJOR FEATURES

- Atypical pigment network
- Blue-white veil
- Atypical vascular pattern
- Irregular streaks / pseudopods
- Irregular dots / globules
- Irregular blotches
- Regression structures

ANCILLARY FEATURES

- Irregular streaks / pseudopods
- Irregular dots / globules
- Irregular blotches
- Regression structures

L4

A Yes

B Yes

C Yes

D —

E —

MAJOR FEATURES

- Atypical pigment network
- Blue-white veil
- Atypical vascular pattern
- Irregular streaks / pseudopods
- Irregular dots / globules
- Irregular blotches
- Regression structures

ANCILLARY FEATURES

- Irregular streaks / pseudopods
- Irregular dots / globules
- Irregular blotches
- Regression structures

L1: The image displays asymmetry, irregular borders, multiple colors (multi-colored), and multiple dermoscopic structures. The calculated TDS score is high (8.90), indicating a high suspicion for melanoma. These features are highly suspicious for melanoma despite the HAM10000 classifier predicting a benign nevus with a low malignancy probability (0.0%).

L2: The image displays asymmetry, irregular borders, multiple colors (multi-colored), and multiple dermoscopic structures. The calculated TDS score is high (7.60), indicating a high suspicion for melanoma. These features are highly suspicious for melanoma despite the HAM10000 classifier predicting a benign nevus with a low malignancy probability (0.0%).

L3: The image displays asymmetry, irregular borders, multiple colors (multi-colored), and multiple dermoscopic structures. The calculated TDS score is high (7.50), indicating a high suspicion for melanoma. These features are highly suspicious for melanoma despite the HAM10000 classifier predicting a benign nevus with a low malignancy probability (0.0%).

L4: The image displays asymmetry, irregular borders, multiple colors (multi-colored), and multiple dermoscopic structures. The calculated TDS score is high (6.90), indicating a high suspicion for melanoma. These features are highly suspicious for melanoma despite the HAM10000 classifier predicting a benign nevus with a low malignancy probability (0.0%).

DIFFERENTIAL DIAGNOSIS

1. Melanocytic nevus (nv)

HIGH

SUPPORTS

- + Asymmetry
- + Irregular borders
- + Multiple colors
- + Multiple dermoscopic structures
- + High TDS score (8.90 - 13 points)

AGAINST

- HAM10000 classifier predicts benign nevus with low malignancy probability (0.0%)

2. Dermatofibroma (df)

LOW

AGAINST

- Features are highly suspicious for melanoma despite classifier prediction

3. Vascular lesion (vasc)

LOW

AGAINST

- Features are highly suspicious for melanoma despite classifier prediction

4. Actinic keratosis / Bowen's (in-situ SCC) (akiec)

LOW

AGAINST

- Features are highly suspicious for melanoma despite classifier prediction

RECOMMENDATIONS

- Urgent excision is recommended due to the high risk level based on the 7-point score and the high malignancy probability from the HAM10000 classifier.

GUIDELINE CITATIONS

[NCCN/AAD guidelines for melanoma assessment and management]

QUANTITATIVE IMAGE ANALYSIS

4 dermoscopy images analysed; per-image metrics below.

Image 1 (melanoma_9608.jpg)

Computed from automated lesion segmentation on the enhanced image

Asymmetry **49.5% area mismatch across principal axes** asymmetric
How lopsided the lesion is around its centre — a higher % is more asymmetric, a melanoma warning sign.

Border irregularity index **1.58**
irregular/notched border
How ragged the edge is. 1.0 is a smooth circle; higher means a more notched, irregular border.

Maximum diameter 168 px
physical mm not available — no calibration in image
Longest width of the lesion. Reported in pixels because the photo carries no real-world scale.

Colour variegation

6 distinct dominant colour cluster(s) (multi-coloured)

Number of distinct colours in the lesion. Three or more colours raises melanoma suspicion.

Lesion covers 11% of the image

Stolz ABCD Total Dermoscopy Score (TDS = $1.3 \cdot A + 0.1 \cdot B + 0.5 \cdot C + 0.5 \cdot D$)

A scored dermoscopy rule: Asymmetry, Border, Colour and Differential structures are weighted into one TDS number. Above ~5.45 is melanoma-suspicious.

A (asymmetry)	2/2 axes
B (border cutoffs)	8/8
C (colours)	6/6
D (structures, estimated)	5/5

TDS = 8.90 -> highly suspicious for melanoma (TDS >5.45)

TDS <4.75 benign · 4.75–5.45 suspicious · >5.45 melanoma-suspicious

automated TDS estimate — A/B measured geometrically, C from colour clustering, D approximated from texture. Decision support only; it does not replace your manual dermoscopic ABCD/7-point assessment.

Image 2 (melanoma_9609.jpg)

Computed from automated lesion segmentation on the enhanced image

Asymmetry **53.0% area mismatch across principal axes** asymmetric

How lopsided the lesion is around its centre — a higher % is more asymmetric, a melanoma warning sign.

Border irregularity index 1.50
irregular/notched border

How ragged the edge is. 1.0 is a smooth circle; higher means a more notched, irregular border.

Maximum diameter 123 px
physical mm not available — no calibration in image
Longest width of the lesion. Reported in pixels because the photo carries no real-world scale.

Colour variegation

6 distinct dominant colour cluster(s) (multi-coloured)

Number of distinct colours in the lesion. Three or more colours raises melanoma suspicion.

Lesion covers 8% of the image

Stolz ABCD Total Dermoscopy Score (TDS = $1.3 \cdot A + 0.1 \cdot B + 0.5 \cdot C + 0.5 \cdot D$)

A scored dermoscopy rule: Asymmetry, Border, Colour and Differential structures are weighted into one TDS number. Above ~5.45 is melanoma-suspicious.

A (asymmetry)	1/2 axes
B (border cutoffs)	8/8
C (colours)	6/6
D (structures, estimated)	5/5

TDS = 7.60 -> highly suspicious for melanoma (TDS >5.45)

TDS <4.75 benign · 4.75–5.45 suspicious · >5.45 melanoma-suspicious

automated TDS estimate — A/B measured geometrically, C from colour clustering, D approximated from texture. Decision support only; it does not replace your manual dermoscopic ABCD/7-point assessment.

Image 3 (melanoma_9610.jpg)

Computed from automated lesion segmentation on the enhanced image

Asymmetry **27.2% area mismatch across principal axes** asymmetric

How lopsided the lesion is around its centre — a higher % is more asymmetric, a melanoma warning sign.

Border irregularity index 1.62
irregular/notched border
How ragged the edge is. 1.0 is a smooth circle; higher means a more notched, irregular border.

Maximum diameter 116 px
physical mm not available — no calibration in image
Longest width of the lesion. Reported in pixels because the photo carries no real-world scale.

Colour variegation

6 distinct dominant colour cluster(s) (multi-coloured)

Number of distinct colours in the lesion. Three or more colours raises melanoma suspicion.

Lesion covers 6% of the image

Stolz ABCD Total Dermoscopy Score (TDS = $1.3 \cdot A + 0.1 \cdot B + 0.5 \cdot C + 0.5 \cdot D$)

A scored dermoscopy rule: Asymmetry, Border, Colour and Differential structures are weighted into one TDS number. Above ~5.45 is melanoma-suspicious.

A (asymmetry) 1/2 axes

B (border cutoffs) 7/8

C (colours) 6/6

D (structures, estimated) 5/5

TDS = 7.50 -> highly suspicious for melanoma (TDS >5.45)

TDS <4.75 benign · 4.75–5.45 suspicious · >5.45 melanoma-suspicious

automated TDS estimate — A/B measured geometrically, C from colour clustering, D approximated from texture. Decision support only; it does not replace your manual dermoscopic ABCD/7-point assessment.

Image 4 (melanoma_9611.jpg)

Computed from automated lesion segmentation on the enhanced image

Asymmetry **16.5% area mismatch across principal axes** asymmetric

How lopsided the lesion is around its centre — a higher % is more asymmetric, a melanoma warning sign.

Border irregularity index 1.63
irregular/notched border
How ragged the edge is. 1.0 is a smooth circle; higher means a more notched, irregular border.

Maximum diameter 175 px
physical mm not available — no calibration in image
Longest width of the lesion. Reported in pixels because the photo carries no real-world scale.

Colour variegation

5 distinct dominant colour cluster(s) (multi-coloured)

Number of distinct colours in the lesion. Three or more colours raises melanoma suspicion.

Lesion covers 17% of the image

Stolz ABCD Total Dermoscopy Score (TDS = $1.3 \cdot A + 0.1 \cdot B + 0.5 \cdot C + 0.5 \cdot D$)

A scored dermoscopy rule: Asymmetry, Border, Colour and Differential structures are weighted into one TDS number. Above ~5.45 is melanoma-suspicious.

A (asymmetry) 1/2 axes

B (border cutoffs) 6/8

C (colours) 5/6

D (structures, estimated) 5/5

TDS = 6.90 -> highly suspicious for melanoma (TDS >5.45)

TDS <4.75 benign · 4.75–5.45 suspicious · >5.45 melanoma-suspicious

automated TDS estimate — A/B measured geometrically, C from colour clustering, D approximated from texture. Decision support only; it does not replace your manual dermoscopic ABCD/7-point assessment.

HAM10000 classifier — per-image predictions below.

Image 1 (melanoma_9608.jpg)

HAM10000 trained classifier (ResNet50, 7-class, temperature-calibrated T=0.12) (val acc 88.9%)

A neural network trained on the HAM10000 dermoscopy dataset to sort lesions into 7 diagnostic classes.

Most likely

Melanocytic nevus (nv) — 100.0%

The single class the trained network scored highest for this lesion.

Estimated malignancy probability

0.0%

sum of melanoma + BCC + AKIEC; operating threshold 5%

Combined probability across the cancerous classes (melanoma + BCC + AKIEC).

Class probabilities

The network's confidence spread across all classes (totals 100%).

Melanocytic nevus (nv)

100.0%

Dermatofibroma (df)

0.0%

Vascular lesion (vasc)

0.0%

Actinic keratosis / Bowen's (in-situ SCC) (akiec)

0.0%

probabilities are calibrated decision support, trained on HAM10000 dermoscopy images; reliability is bounded by that dataset's distribution. Not a substitute for histopathology.

Image 2 (melanoma_9609.jpg)**HAM10000 trained classifier (ResNet50, 7-class, temperature-calibrated T=0.12) (val acc 88.9%)**

A neural network trained on the HAM10000 dermoscopy dataset to sort lesions into 7 diagnostic classes.

Most likely

Melanocytic nevus (nv) — 100.0%

The single class the trained network scored highest for this lesion.

Estimated malignancy probability

0.0%

sum of melanoma + BCC + AKIEC; operating threshold 5%

Combined probability across the cancerous classes (melanoma + BCC + AKIEC).

Class probabilities

The network's confidence spread across all classes (totals 100%).

Melanocytic nevus (nv)

100.0%

Dermatofibroma (df)

0.0%

Vascular lesion (vasc)

0.0%

Actinic keratosis / Bowen's (in-situ SCC) (akiec)

0.0%

probabilities are calibrated decision support, trained on HAM10000 dermoscopy images; reliability is bounded by that dataset's distribution. Not a substitute for histopathology.

Image 3 (melanoma_9610.jpg)**HAM10000 trained classifier (ResNet50, 7-class, temperature-calibrated T=0.12) (val acc 88.9%)**

A neural network trained on the HAM10000 dermoscopy dataset to sort lesions into 7 diagnostic classes.

Most likely

Melanocytic nevus (nv) — 99.9%

The single class the trained network scored highest for this lesion.

Estimated malignancy probability

0.0%

sum of melanoma + BCC + AKIEC; operating threshold 5%

Combined probability across the cancerous classes (melanoma + BCC + AKIEC).

Class probabilities

The network's confidence spread across all classes (totals 100%).

Melanocytic nevus (nv)

99.9%

Dermatofibroma (df)

0.1%

Vascular lesion (vasc)

0.0%

Actinic keratosis / Bowen's (in-situ SCC) (akiec)

0.0%

probabilities are calibrated decision support, trained on HAM10000 dermoscopy images; reliability is bounded by that dataset's distribution. Not a substitute for histopathology.

Image 4 (melanoma_9611.jpg)

HAM10000 trained classifier (ResNet50, 7-class, temperature-calibrated T=0.12) (val acc 88.9%)

A neural network trained on the HAM10000 dermoscopy dataset to sort lesions into 7 diagnostic classes.

Most likely

Melanocytic nevus (nv) — 99.0%

The single class the trained network scored highest for this lesion.

Estimated malignancy probability

0.0%

sum of melanoma + BCC + AKIEC; operating threshold 5%

Combined probability across the cancerous classes (melanoma + BCC + AKIEC).

Class probabilities

The network's confidence spread across all classes (totals 100%).

Melanocytic nevus (nv)	99.0%
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Dermatofibroma (df)	0.8%
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Vascular lesion (vasc)	0.2%
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Actinic keratosis / Bowen's (in-situ SCC) (akiec)	0.0%
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probabilities are calibrated decision support, trained on HAM10000 dermoscopy images; reliability is bounded by that dataset's distribution. Not a substitute for histopathology.

APPROVED

Radiologist: LuminaDx Admin · 6/29/2026, 3:13:57 PM

AI-assisted decision support only. All findings must be reviewed and confirmed by a licensed radiologist before any clinical use.
This report does not constitute a clinical diagnosis.